

Sisi Safe AI

Sisi Safe AI helps women and couples identify safer periods for unprotected sex without contraceptives by providing personalized fertility and cycle guidance.

LIVE APP

sisi-safe-zone.lovable.app

ORCHESTRATION

n8n Workflow Engine

KNOWLEDGE BASE

Supabase + pgvector

AI ENGINE

OpenAI + RAG

Overview

Sisi Safe AI is an AI-powered fertility and menstrual cycle assistant designed to help women and couples better understand fertility patterns, ovulation windows, and safer periods for unprotected sex without contraceptives.

Using personalized cycle information and evidence-based fertility awareness methods, Sisi Safe AI provides easy-to-understand guidance that helps users make informed decisions about family planning, conception timing, and menstrual health.

The platform delivers supportive, culturally relevant guidance tailored to African and Nigerian users through a conversational AI experience — embodied by Sisi, the warm and approachable "Big Sister" persona.

Problem Statement

Many women and couples lack easy access to reliable fertility awareness information and personalized menstrual cycle guidance. Existing tools often provide generic calendar predictions without explaining the reasoning behind fertile and less fertile periods.

As a result, users struggle to:

- Understand their menstrual cycles
 - Identify ovulation and fertile windows
 - Estimate lower-risk periods for unprotected intercourse
 - Track cycle irregularities and fertility patterns
 - Access reproductive guidance in a conversational and relatable manner
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Solution

Sisi Safe AI combines cycle tracking, fertility-awareness principles, and AI-powered guidance to help users:

- Understand menstrual cycle phases
- Predict ovulation windows
- Identify fertile and less fertile days
- Receive personalized cycle insights
- Learn fertility concepts through natural conversations
- Make informed family-planning decisions

The system provides educational guidance and fertility estimates while encouraging users to consult healthcare professionals when necessary.

Target Users

WOMEN

Women seeking personalized fertility awareness guidance, cycle tracking support, and menstrual health education.

COUPLES

Couples who want to plan for pregnancy, understand fertility timing, identify lower-risk periods, and improve reproductive health awareness.

Project Architecture

FRONTEND

A user-friendly conversational interface built with **Lovable**, allowing users to input menstrual cycle information and receive personalized fertility guidance.

ORCHESTRATION LAYER

n8n serves as the workflow engine responsible for:

- Managing user interactions
- Processing cycle data
- Coordinating AI responses
- Integrating external services

KNOWLEDGE BASE

A vector database powered by **Supabase** and **pgvector** stores curated fertility-awareness and reproductive health content, serving as the system's trusted knowledge source.

AI ENGINE

OpenAI models power:

- Natural language understanding
- Personalized fertility guidance
- Educational explanations
- Context-aware conversations

A custom system prompt ensures Sisi maintains a supportive, approachable "Big Sister" personality while remaining evidence-based.



Technical Workflow

1

User Input

Users provide last menstrual period date, average cycle length, period duration, cycle irregularity information, and fertility-related questions.

2

Cycle Analysis

The workflow analyzes cycle information to estimate menstrual phase, ovulation window, fertile days, and lower-fertility periods.

3

Knowledge Retrieval

The n8n agent queries the Supabase vector database to retrieve relevant fertility and reproductive health information.

4

AI Response Generation

The AI combines user cycle data, fertility calculations, and retrieved medical context to generate personalized, easy-to-understand guidance.

5

Response Delivery

Users receive fertility window estimates, safe-period guidance, cycle insights, educational explanations, and appropriate medical disclaimers.

Key Features

Personalized Fertility Guidance

Cycle-specific insights based on user-entered menstrual information rather than generic calendar estimates.

Fertile Window Prediction

Helps users understand when ovulation is likely to occur and identifies days with higher fertility probability.

Lower-Risk Period Identification

Highlights days that may present a lower likelihood of pregnancy based on fertility-awareness principles.

Educational Fertility Coaching

Explains ovulation, menstrual phases, fertility timing, and cycle irregularities in simple, conversational language.

Context-Aware AI (RAG)

Uses Retrieval-Augmented Generation to provide responses grounded in curated fertility and reproductive health information.

Cultural Adaptation

Communicates in a friendly, supportive style that resonates with Nigerian and African users while maintaining accuracy.

Scalable Knowledge Base

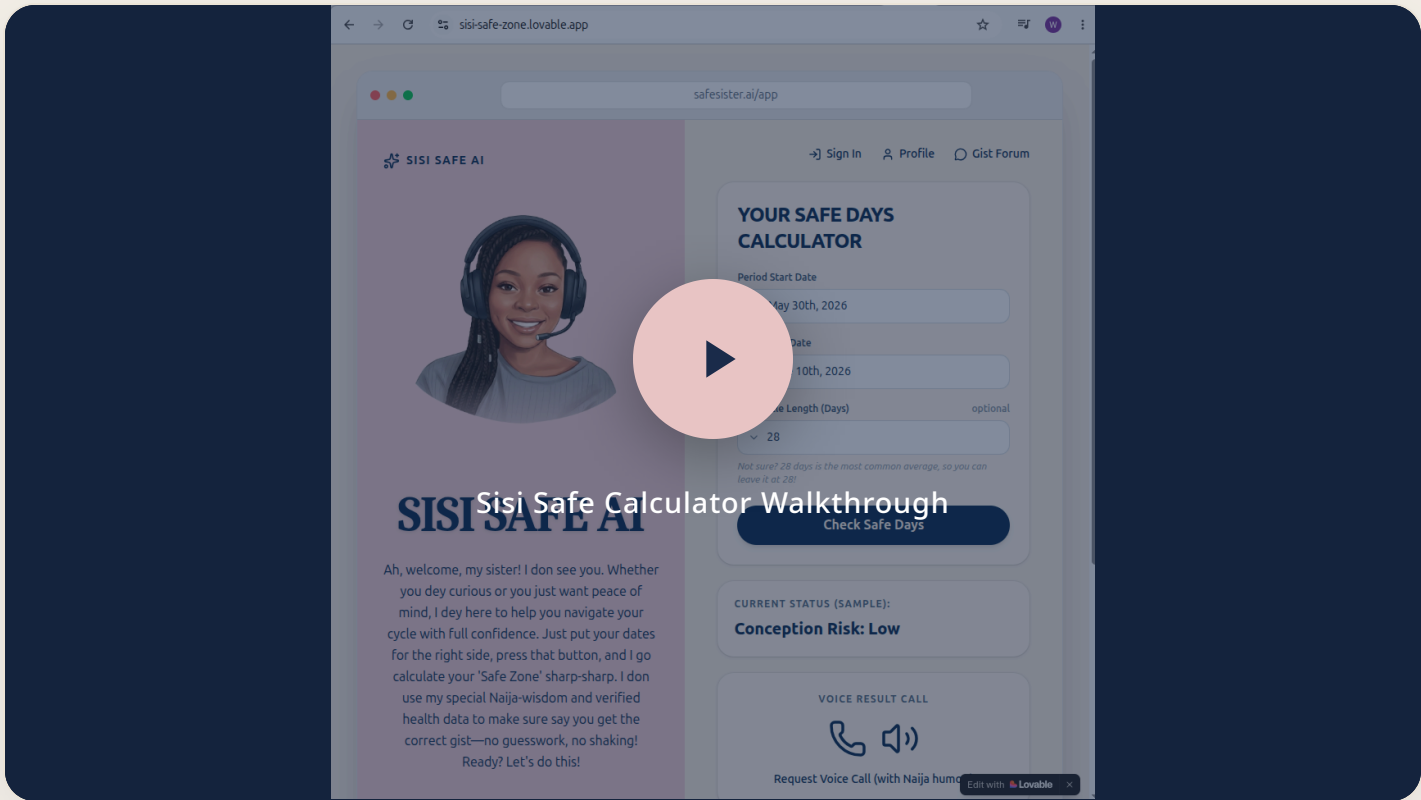
New fertility resources and educational materials can be added without retraining the AI model.

Voice-Assisted Guidance

ElevenLabs text-to-speech delivers fertility guidance in a warm, conversational voice directly in the app.

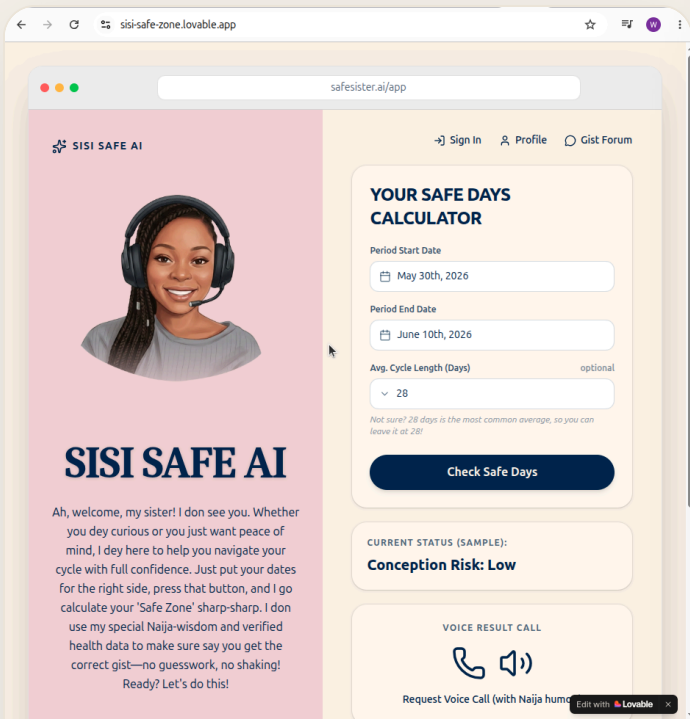
Product Demo

Watch the Sisi Safe Calculator in action — from entering cycle dates to receiving personalized fertility guidance with voice playback.

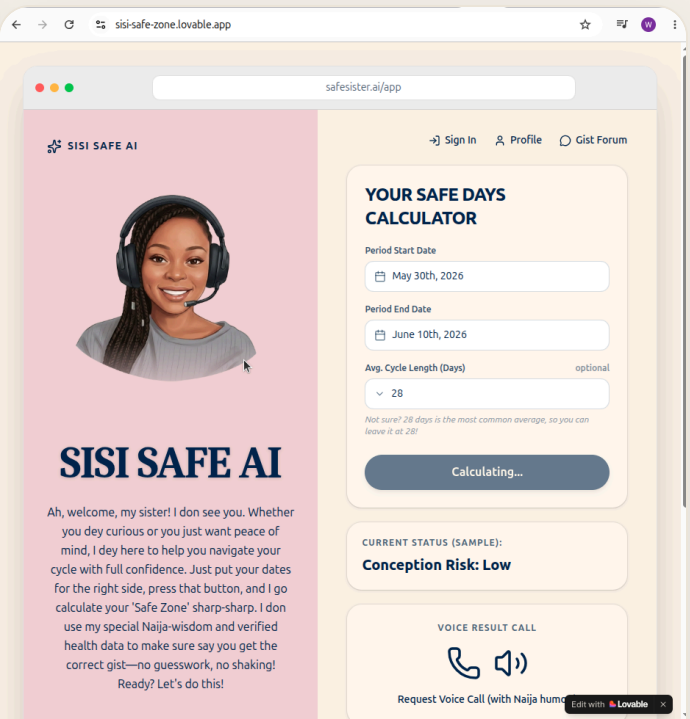


Application & Workflows

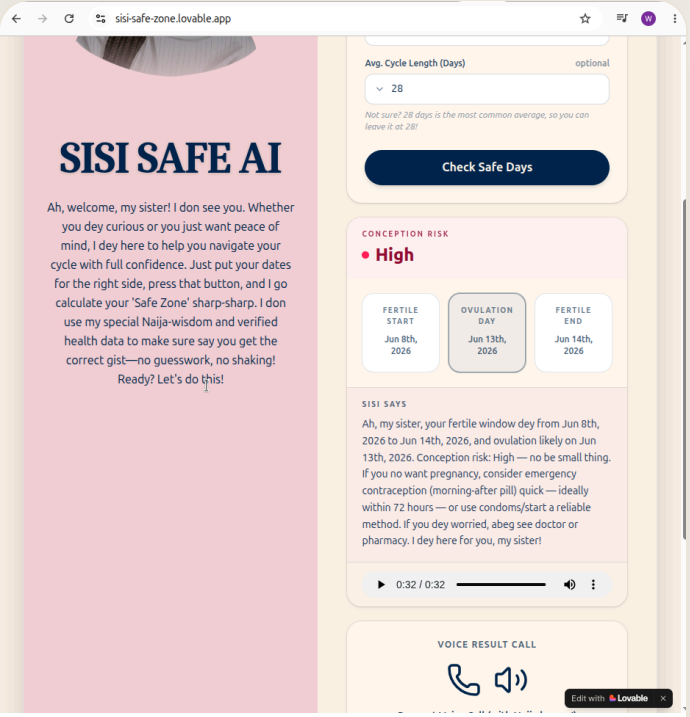
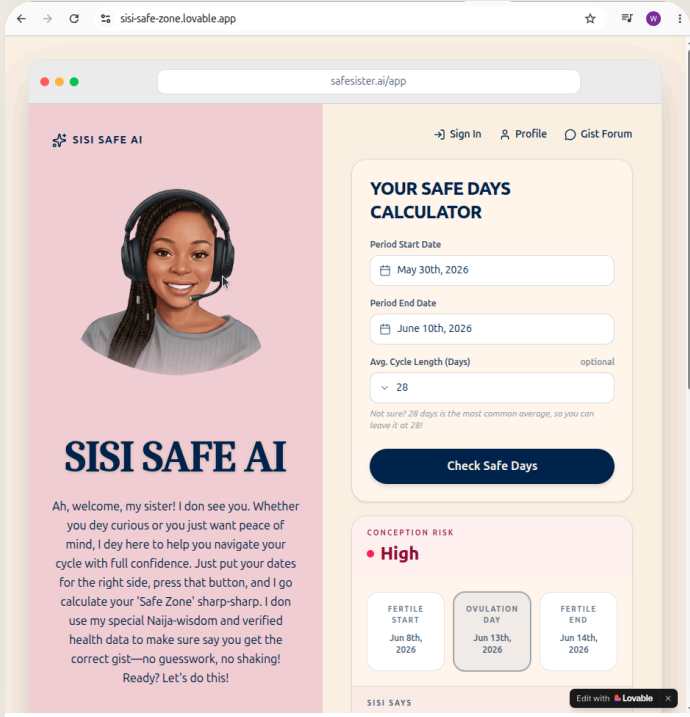
USER INTERFACE



Safe Days Calculator — input form with Sisi's welcoming introduction



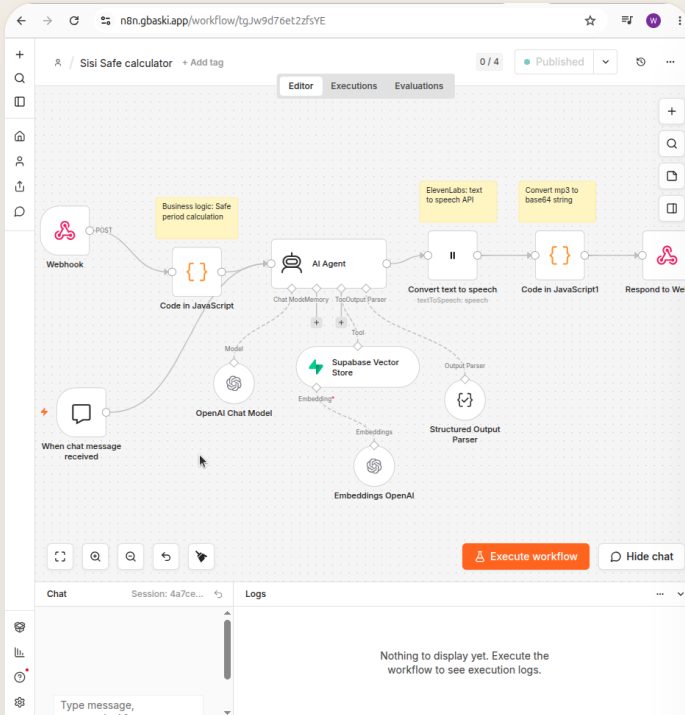
Processing cycle data via n8n backend



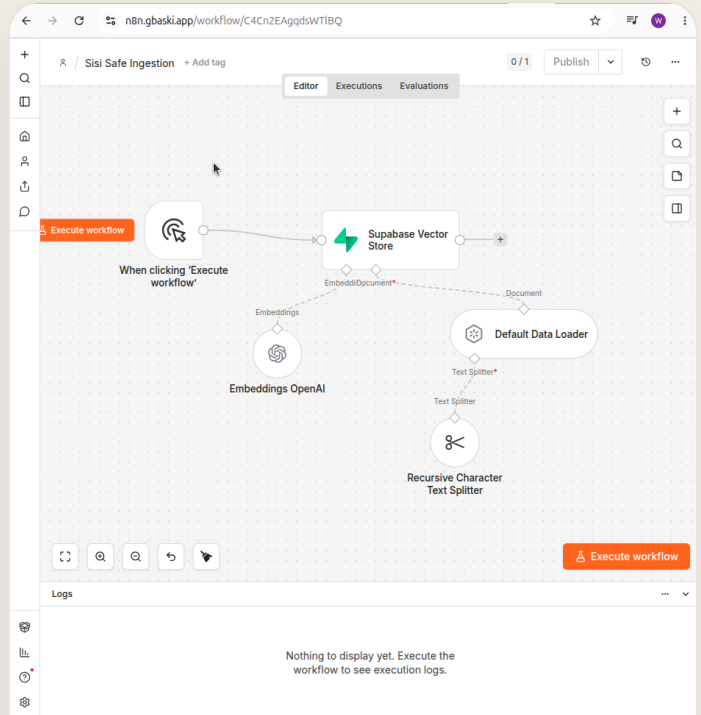
Results showing conception risk and fertile window dates

"Sisi Says" personalized guidance with voice playback

N8N ORCHESTRATION



Sisi Safe Calculator — full AI agent workflow with voice output



Sisi Safe Ingestion — knowledge base vector store pipeline

Voice Integration

To improve accessibility and create a more engaging user experience, Sisi Safe AI supports voice responses. Users can listen to fertility guidance and cycle explanations delivered in a warm, conversational voice.



AI Response

Finalized text guidance from the agent



ElevenLabs TTS

Converts text to natural speech



Base64 Encode

Audio buffer packaged for JSON



HTML5 Player

In-app audio playback for users

Frontend Implementation

The Lovable frontend receives:

- AI text response
- Base64 audio payload
- Audio MIME type

The application automatically renders an HTML5 audio player, allowing users to listen to fertility guidance directly within the conversation interface.

Safety & Medical Disclaimer

Important: Sisi Safe AI provides educational fertility-awareness guidance and cycle-based estimates.

Predictions are **not guaranteed methods of contraception** and should not be considered medical advice. Fertility patterns vary among individuals, especially for users with irregular cycles, hormonal conditions, or underlying health concerns.

Users are encouraged to consult qualified healthcare professionals for medical diagnosis, treatment, or family-planning decisions.

Troubleshooting & Maintenance

ERROR HANDLING

Structured output validation ensures consistent communication between backend services and frontend applications.

DATABASE OPTIMIZATION

The vector search infrastructure uses indexed similarity searches for efficient retrieval from fertility and reproductive health knowledge collections.

MONITORING

Workflow execution, AI responses, and external integrations are monitored through n8n to ensure reliability and performance.

Innovation

WHAT MAKES SISI SAFE AI UNIQUE

Sisi Safe AI combines fertility awareness education, personalized cycle analysis, AI-powered conversations, voice-assisted guidance, and Retrieval-Augmented Generation (RAG) to create an accessible fertility companion that helps women and couples better understand their reproductive cycles and make informed family-planning decisions.

Fertility Awareness Education

Personalized Cycle Analysis

AI Conversations

Voice Guidance

RAG Knowledge Base

Cultural Adaptation